

Ali Shafiee

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[Codeforces](#) | [LinkedIn](#) | [IOI Profile](#) | [Google Scholar](#)

IOI silver medalist and ICPC World Finalist. PhD researcher in probability, stochastic control, and algorithmic game theory.

EDUCATION

Institute of Science and Technology Austria (ISTA)

PhD in Computer Science, advisor [Prof. Krishnendu Chatterjee](#)

Klosterneuburg, Austria

Sep. 2025 – present

Sharif University of Technology

BSc in Computer Engineering

Tehran, Iran

Sep. 2018 – Aug. 2023

COMPETITIVE PROGRAMMING

- **Codeforces**: top 50 worldwide in 2019; International Grandmaster, peak rating 2800 (handle ToTLeS)
- Silver Medal, **International Olympiad in Informatics (IOI)**, 2017: 76th of 300+ contestants from 80 countries
- 33rd Place, **ACM-ICPC World Finals**, Moscow 2021: among 10,000+ teams from 110 countries
- Gold Medal, **ACM-ICPC West Asia Regional Contest**: 1st place 2019, 2nd 2018, 3rd 2020; 100+ teams each year
- Silver Medal, **Asia-Pacific Informatics Olympiad (APIO)**, 2018 and 2017: 20+ participating countries
- Gold Medal, **Iranian National Olympiad in Informatics**, 2016: top 8 of 10,000+ students nationally
- Bronze Medal, **Iranian Mathematical Olympiad**, 2017: top 80 of 20,000+ students nationally

RESEARCH

Complexity and algorithms for **sequential decision-making under uncertainty**: Markov decision processes with partial observation and with adversarial model uncertainty, plus **algorithmic game theory**. Author lists are alphabetical, per convention in the field.

- **Randomise Alone, Reach as a Team**. L. Brice, T. A. Henzinger, A. Montaseri, **A. Shafiee**, K. S. Thejaswini. *CAV 2026*.
- **Qualitative Analysis of ω -Regular Objectives on Robust MDPs**. A. Asadi, K. Chatterjee, E. Goharshady, M. Karrabi, **A. Shafiee**. *AAAI 2026*.
- **Limit-Sure Reachability for Small-Memory Policies in POMDPs is NP-Complete**. A. Asadi, K. Chatterjee, R. Saona, **A. Shafiee**. *UAI 2025*.
- *Under review*: **On the Complexity of Discounted Robust MDPs with L_p Uncertainty Sets** (NeurIPS 2026); Mechanism Design for Multi-Resource Nakamoto Consensus (*WINE 2026*); Quantitative Analysis of ω -Regular Robust MDPs (*AAAI 2027*).

INDUSTRY EXPERIENCE

Quera, Iran's largest competitive-programming and technical-education platform

Contest Manager → Content Manager → Director of CS Education → Business Line Manager

Tehran, Iran

Mar. 2019 – Aug. 2025

- Owned a business line end to end (Oct. 2024 – Aug. 2025); previously led a 40-person division spanning six technical teams (back-end, front-end, data, blockchain, algorithms, fundamentals) and set the education product roadmap.
- Designed and shipped 12+ bootcamps and courses in C++, Python, SQL, Django, Linux/Git, data analysis, machine learning, and deep learning.
- Ran 20+ large-scale programming contests, including sponsored hiring hackathons with corporate partners.

RESEARCH EXPERIENCE

Research Intern & Online Collaborator, ISTA

Advisor: [Prof. Krishnendu Chatterjee](#)

Klosterneuburg, Austria

Apr. 2024 – Dec. 2024

- Proved that limit-sure reachability for small-memory POMDP policies is NP-complete, settling the complexity of a long-open case (UAI 2025).
- Established complexity bounds and algorithms for robust MDPs with ω -regular and discounted objectives (three further papers); implemented and benchmarked the resulting solvers in C++/Python across randomly generated and structured MDP families.

Research Assistant, Sharif University of Technology

Advisor: [Prof. Mohammad Ali Abam](#)

Tehran, Iran

Jun. 2023 – Aug. 2023

- Round- and memory-efficient minimum cut and maximal matching algorithms in the Massive Parallel Computation (MPC) and Adaptive MPC models.

TEACHING & SERVICE

- **Scientific Committee**, ACM-ICPC West Asia Regional Contest (2024 – present); **Host Scientific Committee**, **IOI 2019**, Baku
- **Problem-setter** (2017 – present): Iranian National Olympiad in Informatics, Codecup, and Quera contests
- **Teaching Assistant**, Sharif University: Design of Algorithms, Data Structures & Algorithms, Fundamentals of Programming
- **Olympiad coach** (2017 – 2023): trained 50+ students for the Iranian National Olympiad in Informatics and the IOI

SKILLS

Programming: C++, Python, Rust · **Numerical & ML**: NumPy, SciPy, pandas, Matplotlib, scikit-learn, PyTorch, JAX

Tools: Git, Linux/Bash, SQL, Docker, Jupyter, L^AT_EX · **Mathematics**: probability, stochastic processes, Markov decision processes, optimization, game theory

Languages: Persian (native), English (TOEFL 110/120)